MISSION BRIEFING · SCIENTIST/ENGINEER SC · GREEN LANTERN CORPS EDITION

ISRO CS · NIGHT-BEFORE MISSION CHECKLIST

Wed 28 Oct, 2:00 pm → 1:00 am. 124 checkpoints, and every question pattern from the 2023 and 2025 papers is on board.

🚀 Launch: Thu 29 Oct | Part A · +1 / −⅓ | Part B · no negative | 14 mission phases

Mission rules: tick a box only when you can solve that type of question *without looking*. 23·Q52 = 2023 paper, question 52 (in the app: Study › Past paper, or Quiz › Past papers · subject). Part A: guess only if you can rule out one option. Part B: answer all 15.

2:00 pm – 2:15 pm Warm-up & exam rules ___ / 4

- ☐ Read the **Exam blueprint** chapter: 80 CS Qs (+1 / −⅓) and 15 aptitude Qs (20 marks, no negative)
- ☐ Guess rule: eliminate one option → guess; can't eliminate anything → skip
- ☐ Pack now: admit card, photo ID, 2 black/blue ball pens, watch, water. Know the centre route & reporting time
- ☐ Remember: write AND bubble the booklet SET code on the OMR

2:15 pm – 3:30 pm Operating Systems (8 Qs each year) ___ / 16

- ☐ Process states: which transition is impossible (Waiting → Running) — 25·Q65
- ☐ fork(): parent and child run concurrently / parent waits; 2^n processes; return values — 25·Q64
- ☐ Schedulers: long-term = degree of multiprogramming, medium = swapping, short = fast dispatcher — 23·Q52
- ☐ FCFS by hand with arrival times, avg TAT & WT (also SJF, SRTF, RR) — 25·Q67
- ☐ Response / waiting / turnaround time definitions from a timeline diagram — 25·Q68
- ☐ Banker's algorithm: Need = Max − Alloc, find a safe sequence — 25·Q69
- ☐ Resource-allocation graph → wait-for graph; cycle = deadlock (single instance) — 25·Q66
- ☐ Deadlock conditions + formula $R \geq n(k-1)+1$
- ☐ Page replacement by hand: Optimal, LRU, FIFO (+ Belady's anomaly) — 25·Q70
- ☐ Race condition, critical section, semaphore for producer–consumer — 25·Q63, 23·Q53
- ☐ Priority inversion & priority inheritance — 23·Q49
- ☐ Linux = one-to-one thread model — 23·Q48
- ☐ Trap = software interrupt; illegal memory access → trap to OS — 23·Q51, Q54
- ☐ Segmentation: loader assigns segment numbers; best/first/next fit & locality — 23·Q50, Q9
- ☐ Paging numericals: page-table size, EAT with TLB, disk SSTF/SCAN movement (likely variants)
- ☐ Do: Quiz → Past papers · OS, then 20 OS questions

3:30 pm – 4:30 pm Theory of Computation + Compiler (10 Qs each year) ___ / 12

- ☐ **$S \to aSb \mid T,\ T \to bT \mid b \Rightarrow a^m b^n,\ m < n$** — asked in BOTH years (23·Q57, 25·Q80). Also try S → aaSb | T, T → Tb | a — 23·Q60
- ☐ Read a DFA/NFA diagram → regular expression (ab*, letter(letter|digit)*) — 23·Q55, Q7
- ☐ Minimal DFA state count: 2nd-from-right = b → 4; all non-empty strings → 2 — 23·Q56, 25·Q79
- ☐ Regex equivalences: (a*b)* + (b*a)* = (a+b)* — 25·Q78
- ☐ Closure: regular ∩ CFL is CFL (not always regular); RE not closed under complement — 23·Q58, Q59
- ☐ Strings generated by a small grammar ($a^n c b^n$) — 25·Q33
- ☐ Ambiguous grammar: E → E+E | E×E | id gives exactly 2 trees; ambiguity ⇔ 2 LMD ⇔ 2 RMD — 23·Q8, 25·Q32
- ☐ Compiler phase order; parser receives tokens; yacc builds parsers, lex builds scanners — 23·Q10, 25·Q30, Q31
- ☐ LL(1) = Left-to-right, Leftmost, 1 lookahead; LR family power order — 23·Q11
- ☐ Regex token matching e.g. 0x[0-9A-F]+ (case matters!) — 25·Q28
- ☐ Token counting & which phase catches which error (likely variant)
- ☐ Do: Quiz → Past papers · TOC and · Compiler

4:30 pm – 4:45 pm Break ___ / 1

- [ ] Walk, water, snack — no phone scrolling

4:45 pm – 5:45 pm DBMS & SQL (5–7 Qs) ___ / 10

- [ ] Find the candidate key from FDs, then the highest normal form (partial dependency → not 2NF) — 25·Q48
- [ ] Normal forms: 5NF = join dependency, 4NF = MVD — 23·Q34
- [ ] ER → tables: 1:N FK on the N side; M:N needs its own table; count minimum tables — 25·Q44, Q46
- [ ] select * from R, S = CROSS JOIN; trace an INSERT/SELECT puzzle (Y = $2^x - 1$) — 23·Q31, Q30
- [ ] DELETE with a sub-query on another table — 23·Q32
- [ ] ACID: durability holds always, any failure — 23·Q33
- [ ] Composite index (country, city, street): leftmost-prefix rule — 25·Q45
- [ ] Views restrict sensitive columns; NoSQL = flexible schema — 25·Q43, Q42
- [ ] Transactions under locking: deadlock & rollback scenarios; 2PL, isolation levels — 25·Q47
- [ ] Do: Quiz → Past papers · DBMS, then 15 DBMS questions

5:45 pm – 6:45 pm Computer Networks (6 Qs each year) ___ / 12

- [ ] Sliding window: window (packets) for > X% utilisation = W·Tt / (Tt + RTT) — 23·Q12
- [ ] CSMA/CD: Tt ≥ 2·Tp → min frame or data rate — 23·Q13
- [ ] Header overhead % across layers (5 × 6 B on 270 B → 10%) — 23·Q14
- [ ] Wi-Fi standards & speeds: b 11M, a/g 54M, ac 3.5G, ad 7G — 23·Q15
- [ ] Error detection vs correction (parity/CRC detect; RS/LDPC correct); CRC detects double errors — 23·Q16, Q22
- [ ] Hamming distance decoding: tie ⇒ cannot correct — 25·Q37
- [ ] Subnet masks: are two IPs on the same /26? (router needed) — 25·Q41
- [ ] TCP ideal window = bandwidth × RTT (in bytes) — 25·Q40
- [ ] Blocking ICMP breaks Path MTU discovery — 25·Q39
- [ ] Wrong default gateway = Layer 3 — 25·Q35
- [ ] Ports table (22, 25, 53, 80, 443…), TCP vs UDP header sizes (likely variant)
- [ ] Do: Quiz → Past papers · Networks, then 20 CN questions

6:45 pm – 7:30 pm Digital Logic + Computer Organization (11 / 7 Qs) ___ / 13

- [ ] Gate identities: tied-input NAND/NOR = NOT; NAND of inverted inputs = OR; LOW only when inputs equal = XOR — 23·Q38, Q39, Q40
- [ ] Circuit trace: NOR then NAND — which inputs give LOW — 25·Q55
- [ ] MUX/DEMUX/encoder facts: 1:16 demux → 4 select lines; octal encoder → 3 OR gates; 1-of-32 from 3:8 → 4 — 23·Q35, 25·Q50
- [ ] Full adder = 3 inputs, 2 outputs — 23·Q37
- [ ] Shift-register delay = bits × clock period — 23·Q36
- [ ] Ripple counter: pulses from state 1001 to 0000 (MOD-16 → 7) — 25·Q53
- [ ] Gated D latch transparent when clock HIGH — 25·Q54
- [ ] Binary fraction → decimal; 2's complement multiply in 8 bits — 25·Q49, Q51
- [ ] K-map / SOP simplification (X'Y'Z' + X'YZ' = X'Z') — 25·Q52
- [ ] Vectored interrupt, superscalar, FIFO prefetch, micro-operation — 23·Q18–Q21
- [ ] Reverse Polish (postfix) of A*B+C*D — 23·Q17
- [ ] Pipeline cycles (k + n − 1), cache tag/index/offset, AMAT (likely variants)
- [ ] Do: Quiz → Past papers · Digital Logic and · COA

7:30 pm – 8:15 pm Dinner break ___ / 1

- [ ] Proper meal, short walk. Skim the Formula sheet once while eating if you like

8:15 pm – 9:00 pm Data Structures + Algorithms (12 / 7 Qs) ___ / 12

- [ ] Tree traversals: inorder from a drawn tree; preorder of BST → postorder — 25·Q1, 23·Q4
- [ ] BST inorder predecessor/successor — 23·Q26
- [ ] Diameter of a binary tree is O(n); heap sort O(n log n) — 23·Q3, Q29
- [ ] Substrings of length-m string = m(m+1)/2 — 23·Q46
- [ ] Identify the sort from intermediate arrays (insertion) — 25·Q7
- [ ] MST weight by Kruskal; MST has V − 1 edges — 23·Q1
- [ ] Match Floyd / Dijkstra / Kruskal / Bellman-Ford / multistage to problems — 23·Q2, 25·Q3
- [ ] Greedy = locally optimal choices; Huffman uses frequencies — 25·Q4, Q2
- [ ] Growth-rate ordering ($n \log n < n^{1.5} < n^{\log n} < 2^n$); matrix multiply O(mnp) — 23·Q24, Q45
- [ ] Horner's rule: minimum operations for a polynomial — 23·Q5
- [ ] Radix sort count (n·d·base) & priority-queue answer "any position" — 23·Q25, Q28
- [ ] Do: Quiz → Past papers · DS and · Algorithms

9:00 pm – 9:45 pm Information Security (5 → 9 Qs, rising) ___ / 9

- [ ] Symmetric keys for n people = n(n−1)/2 (16 → 120); n-bit key → 2^n keys — 25·Q8, Q13
- [ ] RSA: given p, q, d find e ($ed \equiv 1 \bmod \varphi$) — 25·Q38
- [ ] Message digest uses no key; X.509 = PKI certificates — 25·Q9, Q15
- [ ] X.800 services: non-repudiation, integrity, confidentiality, access control — 25·Q10
- [ ] IPsec ESP tunnel mode = confidentiality + integrity + hides private IPs — 25·Q36
- [ ] DNSSEC does not encrypt; firewall stops network scanning (not physical attacks); phishing — 25·Q12, Q14, Q11
- [ ] Ping of death = buffer overflow; CAPTCHA vs automated attacks; asymmetric crypto ≠ DoS protection — 23·Q71, Q75, Q74
- [ ] Match encryption / MAC cloning / ARP spoofing / fingerprinting; ingress / egress / NAT / stateful firewall — 23·Q72, Q73
- [ ] Do: Quiz → Past papers · Security, then 15 security questions

9:45 pm – 10:30 pm Discrete & Engineering Maths (7 Qs each year) ___ / 9

- [ ] Determinant = 0 when rows/columns are dependent (spot it, don't compute) — 23·Q41, 25·Q56
- [ ] Eigenvalues of a (block-)triangular matrix — 25·Q57
- [ ] Euler: V − E + F = 2; regular graph E = kV/2; "enclosed regions" excludes the outer face — 23·Q43, 25·Q5
- [ ] Graph facts: degree sum = 2E, girth, eccentricity/radius; number of simple graphs $2^{n(n-1)/2}$ — 23·Q42, Q6
- [ ] Logic: $(p \wedge q) \to r \equiv \neg p \vee \neg q \vee r$; truth table has 2^n rows — 25·Q59, 23·Q47
- [ ] Counting: no-two-consecutive subsets (Fibonacci, 377); $|A^n| = m^n$; words from distinct letters (GAGANYAAN → 24) — 25·Q60–Q62
- [ ] Probability of not drawing a colour; Poisson is discrete — 23·Q44, 25·Q86
- [ ] Triple integral with odd-function cancellation (likely variant of 25·Q58)
- [ ] Do: Quiz → Past papers · Maths

10:30 pm – 10:45 pm Break ___ / 1

- [ ] Stand up, water, eyes off the screen

10:45 pm – 11:30 pm Software Engg + AI/ML + Cloud & Big Data (≈ 20 easy marks) ___ / 11

- [ ] Cyclomatic complexity E − N + 2P; white-box; MC/DC; regression testing — 23·Q79, 25·Q71, 23·Q76, Q77
- [ ] Alpha/beta = acceptance; CMM levels; risk-management order; V-model pairs — 25·Q72–Q74, Q76
- [ ] Fan-in/fan-out statements; Function Point calculation (578 × 1.07) — 25·Q75, Q77
- [ ] ER = static view — 23·Q80
- [ ] AI agent types (no "Unity based"); decision trees do both; Q-learning = RL; hill climbing = search — 25·Q16–Q20
- [ ] FOL connectives; hybrid Bayesian network; K-means = partitional — 23·Q61, Q62, Q65
- [ ] Elasticity; Type-1 vs Type-2 hypervisor; para-virtualization modifies the guest OS — 25·Q21, Q22, 23·Q70
- [ ] Private cloud, cloud bursting, virtualisation vs dual boot — 23·Q67–Q69
- [ ] Big Data Vs (not Visualisation), HDFS purpose, Flink = streaming, NoSQL for unstructured, structured/semi/unstructured — 25·Q23–Q27
- [ ] Cassandra = distributed; JMX = Java management API — 23·Q66, Q64
- [ ] Do: Quiz → Past papers · SE, · AI and · Cloud

11:30 pm – 12:15 am Aptitude (Part B) + Programming ___ / 8

- [ ] Number series with ×n + n pattern (find the wrong term) — 23·Q92
- [ ] Calendar: day of the week via odd days — 25·Q81
- [ ] Syllogisms: either-or pair; "neither" — 25·Q88, Q89
- [ ] Fact-based inference ("must be true") — 23·Q90, 25·Q82, Q83
- [ ] Course of action / survey-bias / workplace judgement — 23·Q91, Q85, Q84
- [ ] Mixtures, weights system, % increase, max rectangle area, cube volume — 25·Q85, Q87, 23·Q94, Q95, 25·Q84
- [ ] Pointer arithmetic (p + 3 with 2-byte int); out-of-bounds ptr[4]; JVM is hardware independent — 23·Q27, 25·Q34, Q29
- [ ] Do: Quiz → Past papers · Aptitude and · Programming

12:15 am – 12:45 am Final sweep of the past papers ___ / 3

- [ ] Quiz → Past papers (all) → finish any you haven't answered
- [ ] Quiz → Wrong: redo every past-paper question you missed today until correct
- [ ] Re-read the starred (★) questions

12:45 am – 1:00 am Formula sheet, then sleep ___ / 2

- [ ] One read of the Last-day formula sheet (tap ? on anything that feels shaky)
- [ ] Alarm set; admit card + ID + pens in the bag; lights out by 1:00

THE OATH · READ IT BEFORE YOU SLEEP

In brightest day, in blackest night,
No evil shall escape my sight.
Let those who worship evil's might,
Beware my power... Green Lantern's light!

Mission complete. Recharge the ring — sleep — then go light up that paper. 🚀